INTELLIGENT CROP RECOMMENDATION SYSTEM BASED ON SOIL PARAMETERS AND IMAGE ANALYSIS

A PROJECT REPORT

Submitted by

MAYURI DEVI S	**811722104088**
MONIKA M	**811722104092**
NANDHINI K	**811722104098**
PRIYADHARSHINI A	**811722104114**

in partial fulfillment of the requirements for the award degree of

Bachelor in Engineering

20CS8504 & PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM - 621112

APRIL 2026

INTELLIGENT CROP RECOMMENDATION SYSTEM BASED ON SOIL PARAMETERS AND IMAGE ANALYSIS

A PROJECT REPORT

Submitted by

MAYURI DEVI S	**811722104088**
MONIKA M	**811722104092**
NANDHINI K	**811722104098**
PRIYADHARSHINI A	**811722104114**

in partial fulfillment of the requirements for the award degree of

Bachelor in Engineering

20CS8504 & PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM - 621112

APRIL 2026

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)
SAMAYAPURAM - 621112

BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**INTELLIGENT CROP RECOMMENDATION SYSTEM BASED ON SOIL PARAMETERS AND IMAGE ANALYSIS**" has been carried out by the students Mayuri Devi S, Monika M, Nandhini K, Priyadharshini A, The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

Date of Viva Voce: ……

Mrs.G.Rajendra Kannammal,M.E.,
SUPERVISOR
Assistant Professor
Department of CSE
K. Ramakrishnan college of technology
Samayapuram-621 112

Mr. R. Rajavarman, M.E., (Ph.D.,)
HEAD OF THE DEPARTMENT
Assistant Professor (Sr. Grade)
Department of CSE
K. Ramakrishnan college of technology
Samayapuram-621 112

INTERNAL EXAMINER **EXTERNAL EXAMNIER**

ABSTRACT

Agriculture plays a vital role in the economy of many countries. Choosing the right crop seed based on soil conditions, climate, and environmental factors is important for improving agricultural productivity. This project proposes a crop seed suggestion system using Machine Learning techniques. The system analyzes parameters such as soil nutrients (nitrogen, phosphorus, potassium), temperature, humidity, rainfall, and soil pH to recommend the most suitable crop seeds for cultivation. Various machine learning algorithms can be used to train the model using historical agricultural data. The trained model predicts the best crop seed that can grow efficiently in the given conditions. This system helps farmers make data-driven decisions, increases crop yield, reduces risk, and promotes sustainable farming practices. The proposed solution can be implemented as a user-friendly application that assists farmers in selecting the appropriate crop seeds for better agricultural productivity.

Keywords: Machine Learning, Crop Recommendation, Precision Agriculture, Soil Nutrient Analysis, Crop Seed Prediction, Agricultural Data Mining, Smart Farming

ACKNOWLGEMENT

We thank our **Dr. N.Vasudevan**, Principal, for his valuable suggestions and support during the course of my research work.

We thank our **Mr. R. Rajavarman**, Head of the Department, Assistant Professor (Sr Grade), Computer Science and Engineering, for his valuable suggestions and support during the course of my research work.

We wish to record my deep sense of gratitude and profound thanks to my Guide **Mrs. G. Rajendra Kannammal, M.E**, Assistant Professor, Computer Science and Engineering for her keen interest, inspiring guidance, constant encouragement with my work during all stages, to bring this thesis into fruition.

We are extremely indebted to our project coordinator **Mrs.M.Mathumathi**, Assistant Professor, Computer Science and Engineering, for her valuable suggestions and support during the course of my research work.

We also thank the faculty and non-teaching staff members of the Computer Science and Engineering, K. Ramakrishnan college of Technology, Samayapuram, for their valuable support throughout the course of my research work.

Finally, we thank our parents, friends and our well wishes for their kind support.

SIGNATURE

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE No.
	ABSTRACT	**iv**
	LIST OF TABLES	**vi**
	LIST OF FIGURES	**viii**
	LIST OF ABBREVIATIONS	**ix**
1	**INTRODUCTION**	**1**
	1.1 DESCRIPTION	1
	1.2 DOMAIN SPECIFICATION	2
	1.2.1 Agriculture Domain	2
	1.2.2 Machine Learning Domain	2
	1.2.3 Deep Learning Domain	3
	1.3 WEB DEVELOPMENT DOMAIN	3
	1.4 DATA SCIENCE DOMAIN	4
	1.4.1 Data Collection	4
	1.4.2 Data Processing	4
	1.4.3 Model Training	4
	1.5 SYSTEM INTEGRATION DOMAIN	5
2	**LITERATURE SURVEY**	**6**
3	**EXISTING SYSTEM**	**16**
4	**PROBLEMS IDENTIFIED**	**18**
5	**PROPOSED SYSTEM**	**20**
6	**SYSTEM REQUIREMENTS**	**22**
	6.1 HARDWARE REQUIREMENTS	22
	6.2 SOFTWARE REQUIREMENTS	23
7	**SYSTEM IMPLEMENTATIONS**	**25**
	7.1 LIST OF MODULES	25
	7.2 MODULES DESCRIPTION	25

	7.2.1 User Interface Module	25
	7.2.2 Soil Input Module	25
	7.2.3 Soil Image Analysis Module	26
	7.2.4 Crop and Fertilizer Recommendation	26
	7.2.5 Database Management Module	26
8	**SYSTEM TESTING**	**27**
	8.1 UNIT TESTING	27
	8.2 INTEGRATION TESTING	27
	8.3 SYSTEM TESTING	27
	8.4 PERFORMANCE TESTING	27
	8.5 SECURITY TESTING	28
	8.6 USABILITY TESTING	28
9	**RESULTS AND DISCUSSION**	**29**
10	**CONCLUSION AND FUTURE WORK**	**30**
	10.1 CONCLUSION	30
	10.2 FUTURE ENHANCEMENTS	30
	APPENDIX A - SOURCE CODE	**33**
	APPENDIX B - SCREENSHOTS	**42**
	REFERENCES	**46**

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.1	Existing System	17
5.1	Proposed System	21
B.1	Home Page	42
B.2	User Registration Page	42
B.3	User Login Page	43
B.4	User Profile Page	43
B.5	Crop Data Entry Page	44
B.6	Crop And Fertilizer Recommendation Page	44
B.7	Soil Image Upload Page	45
B.8	Soil Analysis Result Page	45

LIST OF SYMBOLS AND ABBREVIATIONS

ML	-	Machine Learning
AI	-	Artificial Intelligence
CSRS	-	Crop Seed Recommendation System
N	-	Nitrogen
P	-	Phosphorus
K	-	Potassium
RF	-	Random Forest
DT	-	Decision Tree
DL	-	Deep Learning

CHAPTER 1
INTRODUCTION

1.1 DESCRIPTION

Agriculture is one of the most important sectors that supports the economy and food production across the world. Farmers often face difficulties in selecting the right crop and fertilizer due to lack of proper knowledge about soil nutrients and environmental conditions. Soil properties such as nitrogen, phosphorus, potassium, ph value, humidity, temperature, and rainfall play a major role in determining crop productivity.

With the development of modern technologies like machine learning and deep learning, it has become possible to analyze soil characteristics and provide accurate crop recommendations. These technologies help in processing large amounts of agricultural data and identifying patterns that are useful for predicting the best crops suitable for a particular soil type.

The proposed system crop seed suggestion using machine learning helps farmers by analyzing soil information or soil images to recommend suitable crops and fertilizers. The system accepts soil parameters such as nitrogen, phosphorus, potassium, temperature, humidity, ph, rainfall, and location. It can also analyze soil images using deep learning techniques to determine soil characteristics.

Based on the analysis, the system suggests the most appropriate crop or seed along with the recommended fertilizer. This intelligent system helps farmers make better decisions, improves crop yield, and promotes efficient use of agricultural resources.

In addition, the system provides a digital platform where farmers can easily access recommendations without needing advanced technical knowledge. By simply entering soil data or uploading soil images, farmers can obtain useful

insights about suitable crops and fertilizer usage. This helps reduce the dependency on traditional trial-and-error farming methods.

The use of machine learning models improves the accuracy of predictions by learning from historical agricultural data. As more data becomes available, the system can continuously improve its performance and provide better recommendations. Therefore, this project contributes to the development of smart agriculture systems that support modern farming practices.

1.2 DOMAIN SPECIFICATION

1.2.1 AGRICULTURE DOMAIN

Agriculture is the primary domain of this project. It focuses on improving farming practices by providing recommendations for suitable crops and fertilizers based on soil conditions. Proper crop selection helps increase productivity and reduces the risk of crop failure.

Agriculture depends heavily on soil quality and climatic conditions. Different crops require different nutrient levels and environmental factors to grow effectively. If farmers select crops that are not suitable for the soil type, it can lead to poor yield and wastage of resources. Therefore, analyzing soil characteristics is essential for successful crop production.

This project aims to support farmers by providing scientific recommendations using data-driven techniques. By understanding soil nutrients and environmental conditions, the system helps farmers choose crops that can grow efficiently in their fields.

1.2.2 MACHINE LEARNING DOMAIN

Machine learning is used to analyze soil data and identify patterns from the dataset. By training models using agricultural datasets, the system can predict which crop grows best under specific soil and environmental conditions.

Machine learning algorithms learn from historical data and generate predictions based on input parameters. In this project, the machine learning model analyzes soil nutrients such as nitrogen, phosphorus, potassium, ph level, humidity, temperature, and rainfall. Using these parameters, the system determines the most suitable crop for cultivation.

Machine learning provides several advantages such as improved prediction accuracy, faster data processing, and the ability to handle large datasets. These capabilities make it an effective tool for solving agricultural problems.

1.2.3 DEEP LEARNING DOMAIN

Deep learning techniques are used for soil image analysis. When a soil image is uploaded, the deep learning model processes the image and identifies soil characteristics to recommend appropriate crops and fertilizers.

Deep learning models are capable of extracting important features from images. These features help identify soil texture, color, and composition, which are useful indicators of soil quality. Based on this analysis, the system can estimate soil properties and provide suitable recommendations.

This approach reduces the need for manual soil testing and allows farmers to obtain quick results using simple image inputs.

1.3 WEB DEVELOPMENT DOMAIN

Web development technologies are used to create a user-friendly interface for the system. The frontend is developed using html, css, and javascript, which allows users to enter soil parameters or upload soil images easily. The backend is developed using python, which processes the data and runs the machine learning models.

The web interface acts as a bridge between the user and the machine learning system. It ensures that users can interact with the system without

needing technical expertise. Farmers can access the system through a web browser and submit their soil information for analysis.

The use of web technologies also enables the system to be easily deployed and accessed from different locations. This improves the usability and accessibility of the application.

1.4 DATA SCIENCE DOMAIN

1.4.1 DATA COLLECTION

The dataset used in this project is collected from kaggle and stored in csv format. The dataset includes soil nutrients, environmental parameters, and crop information.

The collected dataset contains several samples representing different soil conditions and the crops suitable for those conditions. These samples are used to train the machine learning model so that it can learn the relationship between soil properties and crop growth.

1.4.2 DATA PROCESSING

Before training the model, the data is cleaned and processed to remove missing values and errors. Data preprocessing helps improve the accuracy and reliability of the machine learning model.

Data preprocessing includes tasks such as removing duplicate data, handling missing values, normalizing numerical data, and converting categorical values into a suitable format. These steps ensure that the dataset is suitable for training the machine learning algorithms.

1.4.3 MODEL TRAINING

Machine learning algorithms are trained using the processed dataset. The trained model learns the relationship between soil parameters and crop growth to provide accurate recommendations.

During the training process, the dataset is divided into training and testing sets. The training data is used to build the model, while the testing data is used to evaluate the performance of the model. After successful training, the model is integrated into the system to provide real-time predictions.

1.5 SYSTEM INTEGRATION DOMAIN

The project integrates machine learning, deep learning, web development, and database technologies. Mysql is used to store data, python is used for model development, and web technologies are used to build the interface.

The integration of these technologies creates a complete intelligent system that can analyze soil data, process images, and generate crop recommendations. The system also stores user inputs and prediction results in the database for future analysis.

This integrated approach ensures efficient data management, accurate predictions, and an easy-to-use platform for farmers. As a result, the system supports smart farming practices and helps improve agricultural productivity.

CHAPTER 2

LITERATURE SURVEY

2.1 A. K. TRIPATHI (2023) – MACHINE LEARNING BASED CROP RECOMMENDATION SYSTEM FOR SMART AGRICULTURE, A. K. TRIPATHI -2023

In this study, a. K. Tripathi (2023) proposed a machine learning-based crop recommendation system designed to help farmers choose the most suitable crops based on soil and climatic conditions. The system uses several input parameters such as nitrogen, phosphorus, potassium levels, soil ph, temperature, rainfall, and humidity. These parameters are collected from agricultural datasets and used to train machine learning models like decision tree and random forest. The research emphasizes the importance of data-driven agriculture where farmers can make informed decisions instead of relying only on traditional knowledge. The model analyzes patterns between soil properties and crop growth to provide accurate recommendations. Experimental results showed that machine learning algorithms significantly improve crop prediction accuracy and help farmers select appropriate seeds for their land conditions. The study concluded that intelligent crop recommendation systems can increase agricultural productivity and reduce crop failure. The system collects important agricultural parameters such as nitrogen, phosphorus, potassium levels, soil ph value, temperature, rainfall, and humidity. These parameters are given as input to machine learning models like Decision Tree and Random Forest to analyze relationships between soil nutrients and crop growth. The model was trained using historical agricultural datasets to improve prediction accuracy. The study highlighted that data-driven crop selection can increase crop yield and reduce farming losses. The proposed system also supports farmers in selecting appropriate seeds based on soil fertility and seasonal conditions. In traditional farming, crop selection is mainly based on farmer experience and seasonal knowledge.

2.2 CROP RECOMMENDATION SYSTEM USING DATA MINING AND MACHINE LEARNING TECHNIQUES ,B. K. RAMESH - 2024

B. K. Ramesh (2024) developed a crop recommendation system that integrates data mining and machine learning techniques to analyze agricultural datasets. The main objective of the research was to support farmers in selecting suitable crops based on soil fertility and environmental conditions. The system collects information about soil nutrients, temperature, humidity, and rainfall. Classification algorithms such as naïve bayes, support vector machine, and decision tree were used to process the data and predict appropriate crops. The study also focused on identifying patterns in historical agricultural data to understand the relationship between soil characteristics and crop yield. The experimental evaluation demonstrated that machine learning models provide reliable predictions and can improve farming efficiency. The proposed system can help farmers reduce production risks and increase crop productivity. The system analyzes agricultural data such as soil nutrients, rainfall patterns, temperature, and humidity to identify the most suitable crop for cultivation. Various classification algorithms were tested to determine the best performing model. The research focused on improving farming efficiency by providing accurate crop suggestions based on scientific analysis rather than traditional farming methods. The study concluded that machine learning models can effectively analyze complex agricultural datasets and support farmers in making informed decisions regarding crop selection and seed planning. The system analyzes multiple environmental parameters including soil nutrients, rainfall, temperature, and humidity to recommend appropriate crops. Random Forest was selected because it handles large datasets effectively and provides higher prediction accuracy compared to traditional classification methods. Agriculture generates large volumes of data related to soil composition, weather conditions, and crop productivity.

2.3 CROP RECOMMENDATION SYSTEM USING RANDOM FOREST ALGORITHM, C. D. PATEL-2023

In this research, c. D. Patel (2023) proposed an intelligent crop recommendation system using the random forest algorithm. The system analyzes multiple agricultural factors including soil nutrients, temperature, rainfall, and humidity to determine the most suitable crop for cultivation. Random forest was selected because it is capable of handling large datasets and provides high prediction accuracy. The model was trained using historical agricultural data collected from different farming regions. The research demonstrated that ensemble learning algorithms like random forest perform better than traditional statistical methods in crop prediction tasks. The system also helps farmers understand the impact of soil fertility and environmental conditions on crop growth. The study concluded that machine learning-based recommendation systems can improve agricultural decision-making and increase crop yield. The system analyzes multiple environmental parameters including soil nutrients, rainfall, temperature, and humidity to recommend appropriate crops. Random Forest was selected because it handles large datasets effectively and provides higher prediction accuracy compared to traditional classification methods. The research involved training the model using agricultural datasets collected from different regions. The results showed that the system can successfully predict suitable crops and assist farmers in selecting the best seeds for cultivation. The study demonstrates the importance of machine learning in improving agricultural productivity. He system collects data related to soil nutrients, rainfall, temperature, and humidity from agricultural datasets. Experimental results showed that the system can recommend crops with high accuracy and efficiency. The research concluded that machine learning-based crop recommendation systems can improve agricultural productivity and promote sustainable farming practices. The proposed model uses agricultural datasets containing soil nutrients, rainfall levels, humidity, and temperature values.

2.4 AUTOMATED MACHINE LEARNING FOR CROP RECOMMENDATION SYSTEMS, D. MADHU SUDHAN REDDY -2024

D. Madhu sudhan reddy (2024) introduced an automated machine learning approach for crop recommendation systems. The proposed model uses automl frameworks that automatically select the best machine learning algorithm and optimize model parameters for accurate prediction. The system processes agricultural data such as soil nutrient levels, rainfall, temperature, and humidity. Multiple algorithms including random forest, gradient boosting, and neural networks were evaluated during the automated model selection process. The research showed that automl simplifies the development of machine learning models and reduces the time required for model training and optimization. Experimental results indicated that automated models achieved high prediction accuracy and improved system performance. The study highlights the potential of automated machine learning in developing intelligent agricultural decision support systems. The study focuses on using automl techniques to automatically select the best machine learning algorithm and optimize its parameters. Agricultural data such as soil nutrient levels, rainfall, temperature, and humidity are used as input features for the model. Several machine learning algorithms were compared to determine the most accurate prediction model. The results demonstrated that automated machine learning significantly improves the efficiency of model development and provides reliable crop recommendations for farmers. The results showed that automated machine learning techniques can improve prediction accuracy while reducing the time required to develop machine learning models. Agricultural datasets containing soil nutrients, rainfall, temperature, humidity, and crop information were used in this research. The study highlights that automl can be an effective solution for developing intelligent agricultural systems that assist farmers in selecting suitable crops.

2.5 DEEP LEARNING BASED CROP RECOMMENDATION SYSTEM USING ENVIRONMENTAL DATA, E. S. KUMAR -2025

In this study, e. S. Kumar (2025) proposed a deep learning-based crop recommendation system that analyzes environmental and soil parameters to suggest suitable crops for cultivation. The model uses artificial neural networks to learn complex relationships between soil nutrients, weather conditions, and crop growth patterns. Input data includes soil ph, nitrogen, phosphorus, potassium levels, rainfall, temperature, and humidity. The deep learning model processes large agricultural datasets and identifies patterns that may not be detected by traditional machine learning algorithms. The results of the study demonstrated improved prediction accuracy compared to conventional classification techniques. The research concluded that deep learning methods can significantly enhance crop recommendation systems and support modern precision agriculture practices. The system uses neural networks to analyze large agricultural datasets containing soil nutrients, weather conditions, and seasonal information. Deep learning models can identify complex relationships between environmental factors and crop growth patterns. The research shows that deep learning methods provide higher prediction accuracy compared to traditional machine learning algorithms. The proposed system helps farmers identify the most suitable crops and seeds for cultivation under specific environmental conditions. The system processes multiple input features such as soil pH, nitrogen, phosphorus, potassium levels, rainfall, humidity, and temperature. Deep learning algorithms are capable of learning hidden patterns in large agricultural datasets. The results demonstrated that deep learning models can provide more accurate crop predictions compared to traditional machine learning algorithms. The study used neural network models to identify complex relationships between soil nutrients and crop growth conditions. The proposed system helps farmers choose crops that are suitable for their land conditions, which leads to increased productivity and better resource management.

2.6 AGRICULTURAL RECOMMENDATION SYSTEM USING DATA MINING TECHNIQUES, M. BENDRE -2023

M. Bendre (2023) developed an agricultural recommendation system based on data mining techniques to assist farmers in selecting suitable crops. The system analyzes large agricultural datasets containing soil characteristics, climatic conditions, and crop information. Data mining algorithms were applied to discover patterns and relationships within the dataset. These patterns were then used to predict appropriate crops for different soil types and environmental conditions. The research highlights the importance of agricultural data analysis in improving crop productivity. The results showed that data mining techniques can effectively identify hidden patterns in agricultural datasets and provide useful recommendations for farmers. The study suggests that integrating data mining with machine learning can further enhance crop recommendation systems. The system focuses on extracting useful knowledge from soil data and climate information to recommend suitable crops. Various data mining algorithms were applied to classify agricultural data and identify patterns related to crop growth. The research highlights that data mining can improve decision-making in agriculture by providing accurate crop recommendations. The proposed system helps farmers reduce crop failure and improve productivity. The system applied classification techniques to recommend suitable crops for specific soil and environmental conditions. The research focuses on extracting useful knowledge from large amounts of agricultural data. Soil characteristics, climate conditions, and historical crop production data were analyzed to identify patterns related to crop growth. The results showed that data mining techniques can effectively analyze agricultural data and support farmers in decision-making. The study emphasizes that data-driven farming can help reduce crop failure and improve agricultural productivity. The system analyzes soil characteristics, climate conditions, and historical crop production data. Classification algorithms were used to identify suitable crops for different soil types and environmental conditions.

2.7 CROP YIELD PREDICTION AND RECOMMENDATION USING MACHINE LEARNING, N. GANDHI -2024

In this research, n. Gandhi (2024) proposed a machine learning model that combines crop yield prediction and crop recommendation. The system analyzes historical agricultural data such as rainfall patterns, soil nutrients, temperature, and humidity. Machine learning algorithms like random forest and linear regression were used to estimate crop yield and suggest suitable crops for cultivation. The study focuses on improving agricultural productivity by helping farmers choose crops that provide higher yield and better profit. The experimental results demonstrated that machine learning techniques can accurately predict crop yield and support effective farming decisions. The research emphasizes the importance of predictive analytics in modern agriculture. The system analyzes historical agricultural data including rainfall, soil nutrients, temperature, and humidity. Machine learning algorithms such as Random Forest and Linear Regression were used to predict crop yield and recommend crops that provide maximum productivity. The study highlights that accurate yield prediction can help farmers select crops that generate higher profit and improve agricultural planning. The system applied classification techniques to recommend suitable crops for specific soil and environmental conditions. The results showed that data mining techniques can effectively analyze agricultural data and support farmers in decision-making. The study emphasizes that data-driven farming can help reduce crop failure and improve agricultural productivity. The research analyzed historical agricultural data including rainfall, soil nutrients, temperature, and humidity. Machine learning algorithms such as Random Forest and Linear Regression were used to build predictive models. The system estimates the expected crop yield and recommends crops that are likely to produce better results. The study shows that combining crop yield prediction with crop recommendation can help farmers select profitable crops. The proposed system can improve agricultural planning and support farmers in maximizing productivity.

2.8 SMART AGRICULTURE SYSTEM FOR CROP RECOMMENDATION USING MACHINE LEARNING, R. SUJATHA-2023

A smart agriculture system that uses machine learning techniques to recommend crops based on soil and environmental parameters. The system collects data such as soil nutrients, temperature, rainfall, and humidity. Machine learning algorithms including decision tree and support vector machine were applied to analyze the dataset. The results showed that the system can accurately predict suitable crops for different soil types and climate conditions. The research highlights that intelligent agricultural systems can help farmers improve crop production and optimize resource usage. The research emphasizes that smart agriculture technologies can improve farming efficiency and reduce the uncertainty involved in crop selection. The system analyzes soil characteristics and environmental conditions such as rainfall, temperature, and humidity. Classification algorithms including Decision Tree and Support Vector Machine were used for crop prediction. The research emphasizes that smart agriculture technologies can improve farming efficiency and reduce the uncertainty involved in crop selection. The system collects data from agricultural sources and processes parameters such as soil nutrients, rainfall, temperature, and humidity. Classification algorithms including Decision Tree and Support Vector Machine were used to analyze the data and generate crop recommendations. The research highlights the importance of integrating technology into agriculture to improve decision-making. The proposed system helps farmers identify suitable crops for their land and reduces the chances of crop failure. The study concluded that machine learning-based agricultural systems can contribute to sustainable farming practices. The research emphasizes that smart agriculture technologies can improve farming efficiency and reduce the uncertainty involved in crop selection. Machine learning algorithms such as Decision Tree and Support Vector Machine were used to analyze the dataset.

2.9 COMPARATIVE STUDY OF MACHINE LEARNING ALGORITHMS FOR CROP RECOMMENDATION, S. PATEL - 2025

S. Patel (2025) conducted a comparative study of different machine learning algorithms used in crop recommendation systems. The research analyzed the performance of algorithms such as k-nearest neighbor, decision tree, random forest, and support vector machine. The dataset used in the study included soil nutrient levels, rainfall, humidity, and temperature data. The evaluation was carried out based on metrics such as prediction accuracy, precision, and recall. The results showed that ensemble models like random forest achieved better performance compared to other algorithms. The study also emphasized the importance of selecting suitable algorithms for agricultural applications. The research provides valuable insights for developing more accurate crop recommendation systems. The research analyzed algorithms such as k-nearest neighbor, decision tree, random forest, and support vector machine. The dataset used in the study included soil nutrient values, rainfall, temperature, and humidity. The performance of each algorithm was evaluated based on accuracy and computational efficiency. The study concluded that ensemble learning models such as random forest provide better performance for crop prediction tasks. The research analyzed algorithms such as k-nearest neighbor, decision tree, random forest, and support vector machine. The dataset included soil nutrients, rainfall, humidity, and temperature data collected from agricultural sources. That ensemble models like random forest achieved better performance compared to other algorithms .the study compared the performance of these algorithms based on accuracy, precision, and computational efficiency. The results showed that ensemble learning models such as random forest provided better performance compared to other algorithms. The research suggests that selecting the appropriate machine learning algorithm is important for improving the effectiveness of crop recommendation systems.

2.10 DEEP LEARNING BASED CROP RECOMMENDATION USING SOIL IMAGE ANALYSIS, V. PANDEY-2023

V. Pandey proposed a deep learning-based crop recommendation system that integrates soil image analysis with environmental data. The system uses convolutional neural networks to analyze soil images and identify soil type. The model also processes environmental parameters such as rainfall, humidity, and temperature to recommend suitable crops. This approach combines image processing and machine learning techniques to improve prediction accuracy. The results showed that integrating soil image analysis with machine learning models can provide more reliable crop recommendations. The research highlights the potential of artificial intelligence in developing advanced agricultural decision-support systems.Deep learning-based crop recommendation system that uses soil image analysis along with environmental data. Convolutional Neural Networks were used to analyze soil images and identify soil type. The system also processes weather data and soil nutrient levels to recommend suitable crops for cultivation. The results showed that combining image analysis with machine learning improves the accuracy of crop prediction systems and supports modern precision agriculture techniques. Soil image analysis with environmental data. The system uses convolutional neural networks to analyze soil images and identify soil types. In addition to image analysis, the model also processes environmental parameters such as rainfall, temperature, and humidity. The combined analysis helps in predicting suitable crops for specific soil conditions. The experimental results showed that the integration of image processing and machine learning significantly improves prediction accuracy. The study demonstrates the potential of artificial intelligence technologies in developing advanced smart farming systems that support farmers in selecting appropriate crops and seeds. The proposed system uses convolutional neural networks to analyze soil images and identify soil type. In addition, environmental data such as rainfall, humidity, and temperature are also used as input features.

CHAPTER 3

EXISTING SYSTEM

The existing system for crop seed suggestion using machine learning is designed to assist farmers and agricultural users in selecting suitable crops based on available agricultural data. In this system, the process begins with data collection, where agricultural information is gathered from various sources such as government agricultural databases, research datasets, and farming records. The collected data typically includes details related to soil nutrients, crop types, climatic conditions, and crop yield statistics. Soil parameters such as nitrogen, phosphorus, potassium, ph level, moisture content, and temperature are important attributes that influence crop growth. In addition, information about crop market prices and historical production data may also be included in the dataset to support better decision-making. The purpose of collecting this large amount of data is to create a strong foundation for building a machine learning model that can analyze patterns and relationships between soil conditions and suitable crops.

Once the data is collected, the next important step in the existing system is data preprocessing. The raw agricultural data may contain missing values, duplicate entries, and inconsistent formats. Therefore, preprocessing techniques are applied to clean and transform the data into a structured format suitable for analysis. This stage involves handling missing values, removing irrelevant attributes, converting categorical values into numerical form, and normalizing the dataset. Data preprocessing improves the accuracy and reliability of the machine learning model by ensuring that only high-quality data is used for training. After preprocessing, the cleaned dataset is ready for further analysis and model development.

After the model is developed, it is integrated into a web application, which acts as the main interface for users. The web application allows farmers

or agricultural users to interact with the system easily. Users can provide input data such as soil parameters, environmental conditions, or crop-related information through the web interface. The application sends this input to the trained machine learning model for processing. At the same time, the system may retrieve additional information from a database, where datasets, model outputs, and agricultural records are stored. The database ensures that the system can efficiently manage and access large amounts of agricultural data whenever required. Finally, based on the user input and the trained machine learning model, the system generates outputs and displays them through the web application. The results may include recommended crop types, analysis of soil parameters, and crop price information that helps farmers understand the market value of the crops. By providing these outputs, the existing system aims to support farmers in making informed agricultural decisions.

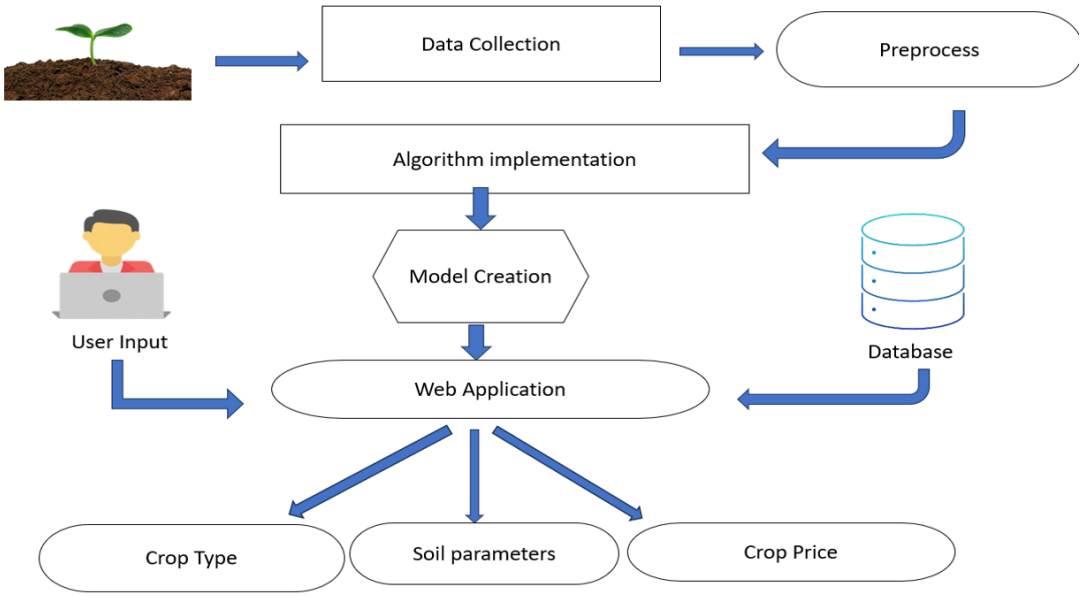

Figure 3.1: Existing System

CHAPTER 4
PROBLEM IDENTIFICATION

Agriculture is the backbone of many developing countries and plays a vital role in food production and economic stability. However, farmers often face significant challenges in choosing the right crops and fertilizers for their land. The growth and productivity of crops depend heavily on soil nutrients, environmental conditions, and proper agricultural practices. If the soil conditions are not suitable for a particular crop, it can lead to poor yield, crop failure, and financial losses for farmers.

One of the major problems in traditional farming is the lack of proper knowledge about soil composition and nutrient levels. Soil contains important nutrients such as nitrogen (n), phosphorus (p), and potassium (k), which are essential for plant growth. In addition, factors such as ph level, humidity, temperature, and rainfall also influence crop production. Many farmers are unaware of how these factors affect crop growth, making it difficult for them to select the most suitable crops for cultivation.

Another challenge is the limited availability of soil testing facilities. Soil testing laboratories are not easily accessible to farmers in rural areas, and the testing process may take a long time and require additional costs. Because of these difficulties, many farmers rely on traditional knowledge or guesswork when selecting crops and fertilizers. This often results in inefficient use of resources and reduced agricultural productivity.

In recent years, the use of digital technologies in agriculture has increased, but many existing systems still require complex data inputs or expert knowledge. Most farmers do not have access to advanced agricultural advisory systems that can analyze soil data and provide accurate crop recommendations.

Furthermore, existing methods usually depend only on numerical soil data and do not consider soil images as an input source.

Another important problem is the lack of integration between different technologies that can support smart agriculture. While machine learning and deep learning techniques have shown great potential in analyzing agricultural data, these technologies are not widely used in practical farming systems. There is a need for an intelligent system that can combine data analysis, image processing, and web-based technologies to assist farmers in making better decisions.

The absence of such systems leads to several problems, including improper crop selection, excessive fertilizer usage, soil degradation, and reduced agricultural productivity. These issues not only affect farmers' income but also impact food security and environmental sustainability.

To overcome these challenges, there is a need for an automated and intelligent system that can analyze soil conditions and provide reliable recommendations for crop cultivation and fertilizer usage. By using machine learning and deep learning techniques, it is possible to develop a system that can analyze soil parameters and soil images efficiently.

The proposed system crop seed suggestion using machine learning aims to address these problems by providing a smart platform where farmers can input soil details such as nitrogen, phosphorus, potassium, temperature, humidity, ph, and rainfall or upload a soil image. The system analyzes the given inputs using trained models and predicts the most suitable crop or seed along with the recommended fertilizer.

This system helps farmers make informed decisions, reduces dependency on traditional trial-and-error methods, and improves overall agricultural productivity.

CHAPTER 5

PROPOSED SYSTEM

The proposed system is designed to provide a more efficient and intelligent solution for crop seed and fertilizer recommendation using machine learning techniques. The system focuses on analyzing soil characteristics accurately and recommending suitable crops and fertilizers based on the soil condition. The process begins with the input stage, where the user provides soil-related information to the system. The user can either upload a soil image captured using a mobile phone or provide soil data such as nutrient levels, ph value, moisture content, and other environmental parameters. This flexibility allows farmers and agricultural users to easily interact with the system without requiring complex technical knowledge. By allowing both image-based input and manual data entry, the system ensures that users from different farming environments can utilize the application effectively.

After receiving the input, the system performs image preprocessing to improve the quality of the uploaded soil image. Image preprocessing techniques such as noise removal, resizing, normalization, and color correction are applied to enhance the clarity of the image and prepare it for further analysis. This step is important because raw images may contain unnecessary variations or distortions that could affect the accuracy of the analysis. Once the image is preprocessed, the system performs feature extraction, where important characteristics of the soil image are identified. These features may include soil color, texture, and other visual properties that help determine the soil type. Extracting these meaningful features allows the system to better understand the physical characteristics of the soil.

In the next stage, the extracted features are matched with a training dataset that contains previously collected soil samples and their corresponding classifications. The proposed system uses machine learning algorithms to

analyze these features and compare them with the dataset to identify patterns and similarities. During this stage, soil analysis is performed to understand the soil composition and its suitability for different crops. The machine learning model evaluates the extracted features and classifies the soil into different categories such as black soil, red soil, sandy soil, or clay soil. This classification helps determine which crops can grow effectively in the identified soil type.

After the soil classification process is completed, the system generates the final output based on the analyzed soil type. The system provides seed and fertilizer recommendations that are suitable for the detected soil condition. These recommendations help farmers choose the most appropriate crops and fertilizers to improve agricultural productivity. The output may also consider environmental factors such as temperature, rainfall, and seasonal conditions to provide better suggestions. By using machine learning and automated soil analysis, the proposed system reduces the need for manual soil testing and provides faster and more accurate recommendations.

Overall, the proposed system improves agricultural decision-making by integrating soil image analysis.

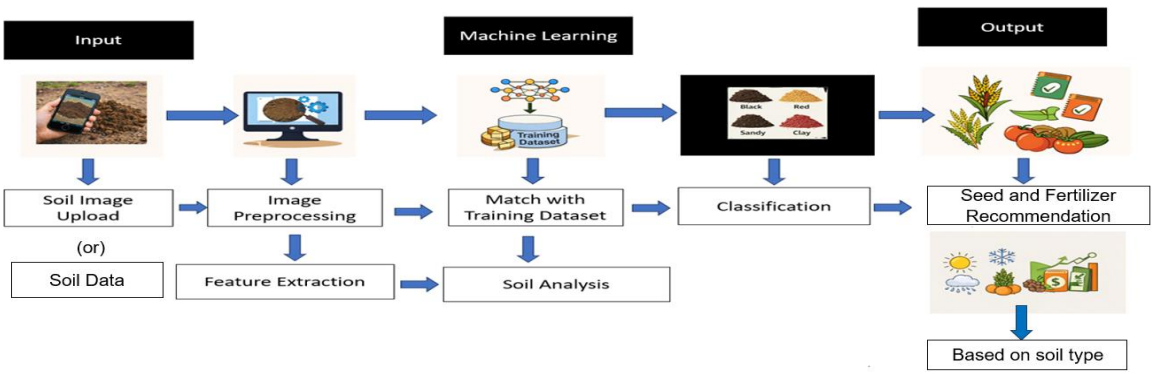

Figure 5.1: Proposed System

CHAPTER 6

SYSTEM REQUIREMENTS

System requirements define the hardware and software components necessary for developing and running the proposed system. These requirements ensure that the system operates efficiently and supports all functionalities such as data input, data processing, machine learning model execution, image analysis, and output generation.

The crop seed suggestion using machine learning system requires both hardware and software resources to perform operations such as collecting soil data, analyzing datasets, training machine learning models, and providing crop and fertilizer recommendations. Proper system requirements help ensure that the system runs smoothly without performance issues.

6.1 HARDWARE REQUIREMENTS

Hardware requirements refer to the physical devices required to develop and run the application. Since the proposed system mainly performs data processing and machine learning operations, it requires a computer system with moderate specifications.

The following hardware components are required for the development and execution of the system:

Processor:

A system with an intel core i3 processor or higher is recommended. A faster processor helps in handling data processing and running machine learning models efficiently.

Random access memory (ram):

At least 4 gb of ram is required to run the development tools, database, and machine learning libraries simultaneously. For better performance, 8 gb ram is recommended.

Hard disk storage:

A minimum of 500 gb hard disk storage is required to store datasets, project files, system libraries, and database information.

Monitor:

A display monitor with a minimum size of 15 inches is required to view the application interface and development environment comfortably.

Input devices:

Standard input devices such as a keyboard and mouse are required for system interaction and data entry.

Internet connection:

An internet connection is necessary for downloading datasets, installing libraries, and accessing online resources.These hardware components provide a stable environment for developing and executing the proposed machine learning application.

6.2 SOFTWARE REQUIREMENTS

Software requirements include the programming languages, development tools, frameworks, and libraries used to build the system. The proposed system integrates web development technologies with machine learning and database systems.

The following software components are used in the development of the project:

Operating system:

The system can run on windows 10 or windows 11 operating systems. These platforms provide compatibility with development tools and machine learning libraries.

Programming language:

Python is used as the main programming language for implementing machine learning algorithms and backend processing.

Frontend technologies:

Html, css, and javascript are used to design the user interface of the web application. These technologies allow users to easily enter soil parameters or upload soil images.

Development environment:

Pycharm community edition 2019.3 is used as the integrated development environment (ide) for writing and managing the python code.

Database management system:

Mysql is used to store user inputs, system data, and prediction results. It helps in organizing and managing data efficiently.

Web server:

Wampserver 2.0 is used to host and manage the web application locally during development.

Dataset source:

The dataset used in the project is collected from kaggle and stored in csv file format. The dataset contains soil parameters and corresponding crop recommendations.

Machine learning libraries:

Several python libraries are used for data processing and model development. These include numpy for numerical operations, pandas for data manipulation, scikit-learn for machine learning algorithms, and tensorflow or keras for deep learning models.

These software tools and technologies work together to build a complete intelligent system capable of analyzing soil data and providing crop recommendations.

Another challenge is the limited availability of soil testing facilities. Soil testing laboratories are not easily accessible to farmers in rural areas, and the testing process may take a long time and require additional costs. Because of these difficulties, many farmers rely on traditional knowledge or guesswork when selecting crops and fertilizers. This often results in inefficient use of resources and reduced agricultural productivity.

To overcome these challenges, there is a need for an automated and intelligent system that can analyze soil conditions and provide reliable recommendations for crop cultivation and fertilizer usage. By using machine learning and deep learning techniques, it is possible to develop a system that can analyze soil parameters and soil images efficiently.

The proposed system crop seed suggestion using machine learning aims to address these problems by providing a smart platform where farmers can input soil details such as nitrogen, phosphorus, potassium, temperature, humidity, ph, and rainfall or upload a soil image. The system analyzes the given inputs using trained models and predicts the most suitable crop or seed along with the recommended fertilizer.

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 LIST OF MODULES

The proposed system is divided into several modules to perform the crop recommendation process efficiently. Each module performs a specific task in the system.

1. User Interface Module
2. Soil Details Input Module
3. Soil Image Analysis Module
4. Crop and Fertilizer Recommendation Module
5. Database Management Module

7.2 MODULES DESCRIPTION

7.2.1 USER INTERFACE MODULE

The User Interface module allows the user or farmer to interact with the system. The interface is developed using HTML, CSS and JavaScript.

Through this interface, users can provide soil details such as location, nitrogen, phosphorus, potassium, temperature, humidity, pH and rainfall.

Users can also upload soil images to analyze the soil condition. The interface is designed to be simple and easy to use so that farmers can access the system without technical knowledge.

7.2.2 SOIL DETAILS INPUT MODULE

This module collects soil parameter values from the user. The parameters include:

- Location
- Nitrogen level
- Phosphorus level
- Potassium level
- Temperature
- Humidity
- pH value
- Rainfall

These inputs are processed using Machine Learning algorithms to determine the most suitable crop and fertilizer. The dataset used for training is obtained from Kaggle and stored in CSV format.

7.2.3 SOIL IMAGE ANALYSIS MODULE

This module analyzes soil images uploaded by the user. The image is processed using Deep Learning techniques to identify soil characteristics.

The system extracts important features from the soil image and compares them with the trained dataset. Based on this analysis, the system predicts the soil condition and determines suitable crops.

7.2.4 CROP AND FERTILIZER RECOMMENDATION MODULE

This module is responsible for recommending the best crop and fertilizer.The recommendation is generated using both Machine Learning and Deep Learning models. The trained model analyzes the soil parameters and soil image features to predict the most appropriate crop and fertilizer.The result is displayed to the user through the web interface.

7.2.5 DATABASE MANAGEMENT MODULE

The system uses MySQL database to store and manage data. The dataset collected from Kaggle is stored in CSV format and imported into the database.

The database stores soil parameters, crop details, fertilizer information and prediction results. This helps the system retrieve data efficiently and improves overall performance.

To overcome these challenges, there is a need for an automated and intelligent system that can analyze soil conditions and provide reliable recommendations for crop cultivation and fertilizer usage. By using machine learning and deep learning techniques, it is possible to develop a system that can analyze soil parameters and soil images efficiently.

The proposed system crop seed suggestion using machine learning aims to address these problems by providing a smart platform where farmers can input soil details such as nitrogen, phosphorus, potassium, temperature, humidity, ph, and rainfall or upload a soil image. The system analyzes the given inputs using trained models and predicts the most suitable crop or seed along with the recommended fertilizer.

One of the major problems in traditional farming is the lack of proper knowledge about soil composition and nutrient levels. Soil contains important nutrients such as nitrogen (n), phosphorus (p), and potassium (k), which are essential for plant growth.

CHAPTER 8

SYSTEM TESTING

System testing is an important phase of software development. It ensures that the system works correctly and produces accurate results. Different types of testing are performed to verify the functionality and performance of the system.

8.1 UNIT TESTING

Unit testing is performed to test individual modules of the system. Each module such as input processing, image analysis and recommendation modules are tested separately.This testing ensures that every module performs its intended function correctly.

8.2 INTEGRATION TESTING

Integration testing is carried out after unit testing. In this phase, all modules are combined and tested together.The system verifies whether the input module, machine learning model and database work together properly without errors.

8.3 SYSTEM TESTING

System testing checks the complete system functionality. The entire application is tested with real input values such as soil parameters and soil images.The system ensures that the crop and fertilizer recommendations are generated correctly based on the given inputs.

8.4 PERFORMANCE TESTING

Performance testing evaluates how efficiently the system works under different conditions.The system checks how quickly it processes soil data and

images and provides recommendations. This testing ensures the system performs well even when multiple users access it.

8.5 SECURITY TESTING

Security testing ensures that the system protects user data.The system verifies that unauthorized users cannot access the database or modify the data. Proper validation is implemented for user inputs and uploaded images.

8.6 USABILITY TESTING

Usability testing ensures that the system is easy to use for farmers and general users.The user interface is tested to make sure users can easily enter soil details, upload images and view crop recommendations without confusion.

This stage involves handling missing values, removing irrelevant attributes, converting categorical values into numerical form, and normalizing the dataset.

Data preprocessing improves the accuracy and reliability of the machine learning model by ensuring that only high-quality data is used for training. After preprocessing, the cleaned dataset is ready for further analysis and model development.

CHAPTER 9

RESULTS AND DISCUSSION

The crop seed suggestion using machine learning system was successfully developed and tested using soil data and soil image inputs. The system analyzes soil parameters such as nitrogen, phosphorus, potassium, temperature, humidity, ph value, rainfall, and location to predict the most suitable crop and fertilizer.

During the testing phase, different soil datasets were provided as input to evaluate the performance of the machine learning model. The system processed the input data and generated crop recommendations based on the trained model. The results showed that the system was able to predict suitable crops accurately based on soil conditions.

In addition to soil data input, the system also supports soil image analysis using deep learning techniques. When a soil image is uploaded, the deep learning model processes the image and extracts important features such as soil texture and color. Based on this analysis, the system provides crop and fertilizer recommendations.

The web-based interface makes the system easy to use. Users can enter soil details or upload soil images through the interface and obtain results quickly. The prediction results are displayed clearly, allowing users to understand which crop and fertilizer are suitable for their soil.

The testing results demonstrate that the proposed system improves decision-making in agriculture by providing accurate and timely recommendations. The integration of machine learning, deep learning, and web technologies makes the system efficient and user-friendly.

Overall, the system successfully meets its objective of helping farmers select suitable crops and fertilizers based on soil conditions.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

The crop seed suggestion using machine learning project was developed to assist farmers in selecting suitable crops and fertilizers based on soil conditions. The system uses machine learning algorithms to analyze soil parameters such as nitrogen, phosphorus, potassium, temperature, humidity, ph, and rainfall.The system also incorporates deep learning techniques to analyze soil images and identify soil characteristics. By combining both data-based and image-based analysis, the system provides accurate recommendations for crop cultivation.The developed web-based platform allows users to easily input soil details or upload soil images. The system processes the input data and predicts the most appropriate crop and fertilizer for the given soil conditions.This project demonstrates how modern technologies such as machine learning, deep learning, and web development can be applied to solve real-world agricultural problems. The system helps farmers make better decisions, increases crop productivity, and promotes efficient use of resources.Therefore, the proposed system provides a practical and intelligent solution for improving agricultural practices and supporting sustainable farming.

10.2 FUTURE ENHANCEMENTS

Although the proposed system provides useful crop recommendations, several improvements can be made in the future to enhance its functionality and performance.In future, the system can be improved by integrating real-time weather data to provide more accurate crop predictions. Weather conditions such as rainfall, temperature, and humidity significantly affect crop growth, and incorporating real-time weather information will improve the reliability of recommendations.

Another enhancement is the integration of mobile applications so that farmers can easily access the system using smartphones. A mobile-friendly application will make the system more accessible, especially for farmers in rural areas.The system can also be expanded to support multiple languages to make it easier for farmers from different regions to use the application.Additionally, more advanced deep learning models can be implemented to improve soil image analysis and provide more accurate predictions. By using larger datasets and more sophisticated algorithms, the system can further improve its performance.Future versions of the system can also include pest detection and crop disease prediction, which will help farmers take preventive measures and protect their crops.With these enhancements, the system can become a complete smart agriculture platform that supports farmers in making informed agricultural decisions.

APPENDIX – A

SOURCE CODE

soilmodel.py

```
import warnings

import tensorflow as tf

from tensorflow.keras.preprocessing.image import ImageDataGenerator

from tensorflow.keras.optimizers import RMSprop

from sklearn.metrics import classification_report

warnings.filterwarnings('ignore')

# Part 1 - Building the CNN

batch_size = 32

# All images will be rescaled by 1./255

train_datagen = ImageDataGenerator(rescale=1/255)

# Flow training images in batches of 128 using train_datagen generator

# Note: The comment says 128, but the variable batch_size is 32

train_generator = train_datagen.flow_from_directory(

    'SData',  # This is the source directory for training images

    target_size=(200, 200),  # All images will be resized to 200 x 200

    batch_size=batch_size,

    # Specify the classes explicitly

    classes=['AlluvialSoil', 'ClaySoil', 'RedSoil', 'YellowSoil'],
```

```
    # Since we use categorical_crossentropy loss, we need categorical labels
    class_mode='categorical'
)
# Model Definition
model = tf.keras.models.Sequential([
# Note the input shape is the desired size of the image 200x200 with 3 bytes color
    # The first convolution
    tf.keras.layers.Conv2D(16, (3,3), activation='relu', input_shape=(200, 200, 3)),
    tf.keras.layers.MaxPooling2D(2, 2),
    # The second convolution
    tf.keras.layers.Conv2D(32, (3,3), activation='relu'),
    tf.keras.layers.MaxPooling2D(2,2)
# The third convolution
    tf.keras.layers.Conv2D(64, (3,3), activation='relu'),
    tf.keras.layers.MaxPooling2D(2,2),
# The fourth convolution
    tf.keras.layers.Conv2D(128, (3,3), activation='relu'),
    tf.keras.layers.MaxPooling2D(2,2),
# The fifth convolution
    tf.keras.layers.Conv2D(256, (3,3), activation='relu'),
```

```
    tf.keras.layers.MaxPooling2D(2,2),
 # Flatten the results to feed into a dense layer
    tf.keras.layers.Flatten(),
# 512 neuron in the fully-connected layer
    tf.keras.layers.Dense(512, activation='relu'),
# 4 output neurons for 4 classes with the softmax activation
    tf.keras.layers.Dense(4, activation='softmax')
])


model.summary()
# Compilation and Training
early = tf.keras.callbacks.EarlyStopping(monitor='val_loss', patience=5)
model.compile(
    loss='categorical_crossentropy',
    optimizer=RMSprop(learning_rate=0.001),
    metrics=['accuracy']
)
total_sample = train_generator.n
n_epochs = 50
history = model.fit(
    train_generator,
```

```
    steps_per_epoch=int(total_sample/batch_size),
    epochs=n_epochs,
    verbose=1
)
```

app.py

```
import os
import numpy as np
from flask import Flask, request, render_template
from tensorflow.keras.models import load_model
from tensorflow.keras.preprocessing import image
# Initialize Flask app
app = Flask(__name__)

# Load the trained model
# Ensure the model file 'soil_model.h5' exists in your project directory
model = load_model('soil_model.h5')
# Define the classes
classes = ['AlluvialSoil', 'ClaySoil', 'RedSoil', 'YellowSoil']
def model_predict(img_path, model):
    # Loading and resizing image to match model input
```

```
    img = image.load_img(img_path, target_size=(200, 200))
   # Preprocessing the image
    x = image.img_to_array(img)
    x = x / 255
    x = np.expand_dims(x, axis=0)
    # Making prediction
    preds = model.predict(x)
    index = np.argmax(preds, axis=1)[0]
    result = classes[index]
    return result
@app.route('/', methods=['GET'])
def index():
    # Main page
    return render_template('index.html')
@app.route('/predict', methods=['GET', 'POST'])
def upload():
    if request.method == 'POST':
        # Get the file from post request
        f = request.files['file']
# Save the file to ./uploads
        basepath = os.path.dirname(__file__)
```

```
        file_path = os.path.join(basepath, 'uploads', f.filename)
# Create uploads folder if it doesn't exist
        if not os.path.exists(os.path.join(basepath, 'uploads')):
            os.makedirs(os.path.join(basepath, 'uploads'))
            f.save(file_path)
# Make prediction
        result = model_predict(file_path, model)
        return result
    return None
if __name__ == '__main__':
    app.run(debug=True)
```

cropDeployment.py

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
import pickle
from sklearn.model_selection import train_test_split
```

```
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import classification_report
warnings.filterwarnings('ignore')
# --- Loading and Cleaning Dataset ---
df = pd.read_csv('./Crop_recommendation.csv')
def clean_dataset(df):
    assert isinstance(df, pd.DataFrame), "df needs to be a pd.DataFrame"
    df.dropna(inplace=True)
    # Removing rows with nan or infinity values
    indices_to_keep = ~df.isin([np.nan, np.inf, -np.inf]).any(axis=1)
    return df[indices_to_keep].astype(np.float64, errors='ignore')
# Map labels to numeric values
label_mapping = {
    'rice': 0, 'maize': 1, 'chickpea': 2, 'kidneybeans': 3,
    'pigeonpeas': 4, 'mothbeans': 5, 'mungbean': 6, 'blackgram': 7,
    'lentil': 8, 'pomegranate': 9, 'banana': 10, 'mango': 11,
    'grapes': 12, 'watermelon': 13, 'muskmelon': 14, 'apple': 15,
    'orange': 16, 'papaya': 17, 'coconut': 18, 'cotton': 19,
    'jute': 20, 'coffee': 21
}
df['label'] = df['label'].map(label_mapping)
```

```
# Visualization

sns.countplot(x=df['label'])

plt.show()

# --- Model Building ---

# Replacing 0 values with NaN for specific columns if necessary

cols_to_fix = ['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall']

df[cols_to_fix] = df[cols_to_fix].replace(0, np.nan)

df.dropna(inplace=True)

# Split Features and Target

X = df.drop(columns='label')

y = df['label']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20,
random_state=0)

# MLP Classifier

classifier = MLPClassifier(random_state=0)

classifier.fit(X_train, y_train)

# Predictions and Reporting

y_pred = classifier.predict(X_test)

print(classification_report(y_test, y_pred))

print("Accuracy on training set: {:.2f}".format(classifier.score(X_train,
y_train)))

print("Accuracy on test set: {:.3f}".format(classifier.score(X_test, y_test)))
```

```
# --- Saving and Loading the Model ---
filename = 'crop-prediction-rfc-model.pkl'
pickle.dump(classifier, open(filename, 'wb'))
# Load model for a single prediction test
loaded_model = pickle.load(open(filename, 'rb'))
# Example prediction with sample data
# Order: N, P, K, temperature, humidity, ph, rainfall
sample_data = np.array([[85, 58, 41, 21.77, 80.31, 7.03, 226.65]])
my_prediction = loaded_model.predict(sample_data)
if my_prediction == 0:
    Answer = 'rice'
else:
    Answer = 'No-Crop-Found'
print(f"Prediction: {Answer}")
```

APPENDIX B-SCREENSHOTS

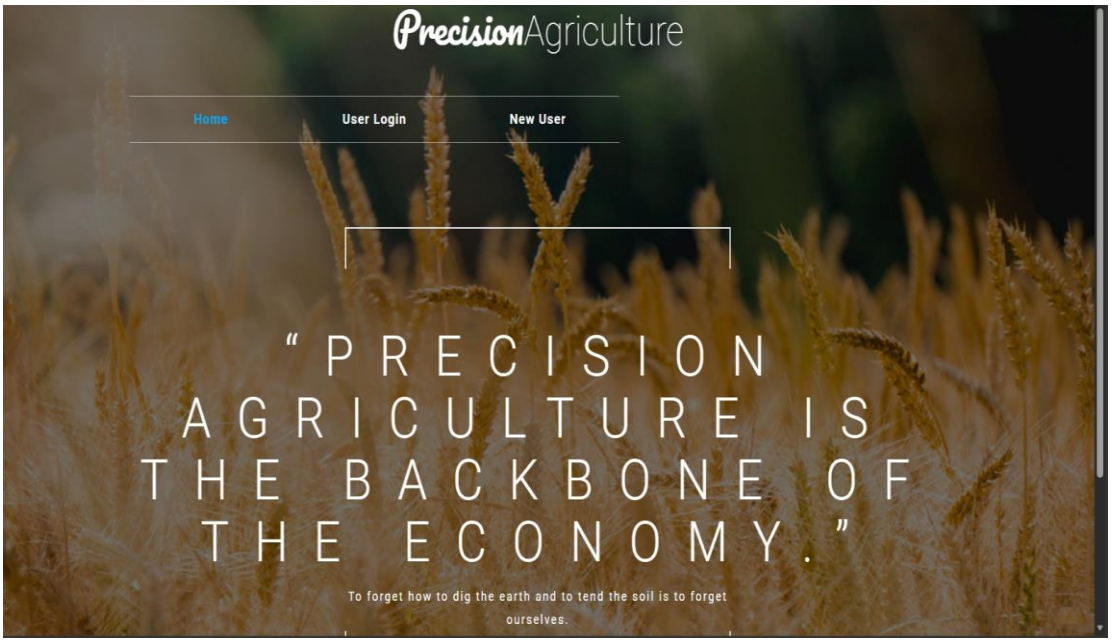

Figure B1. Home page

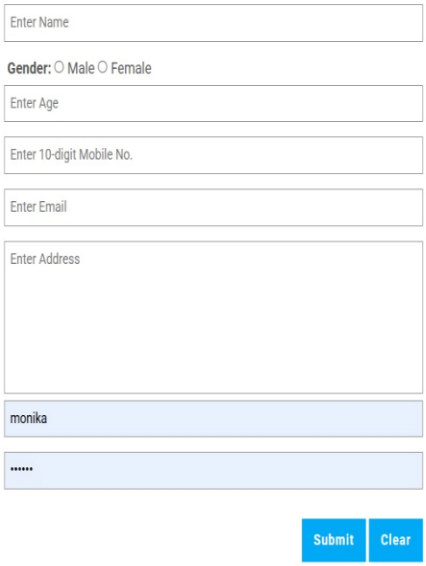

Figure B2. User Registration Page

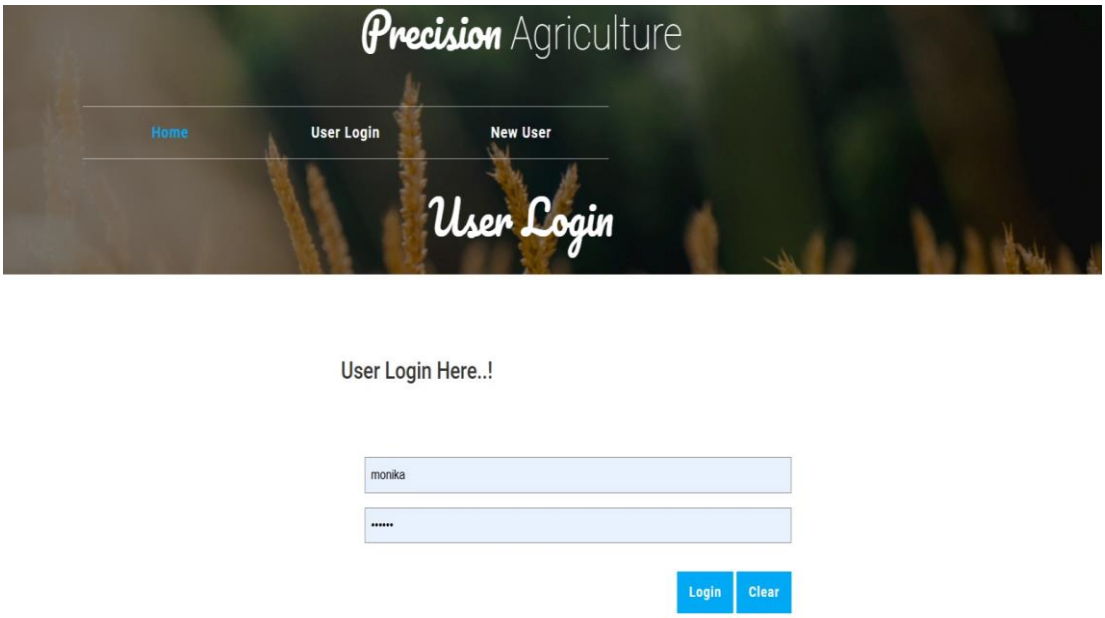

Figure B3. User Login Page

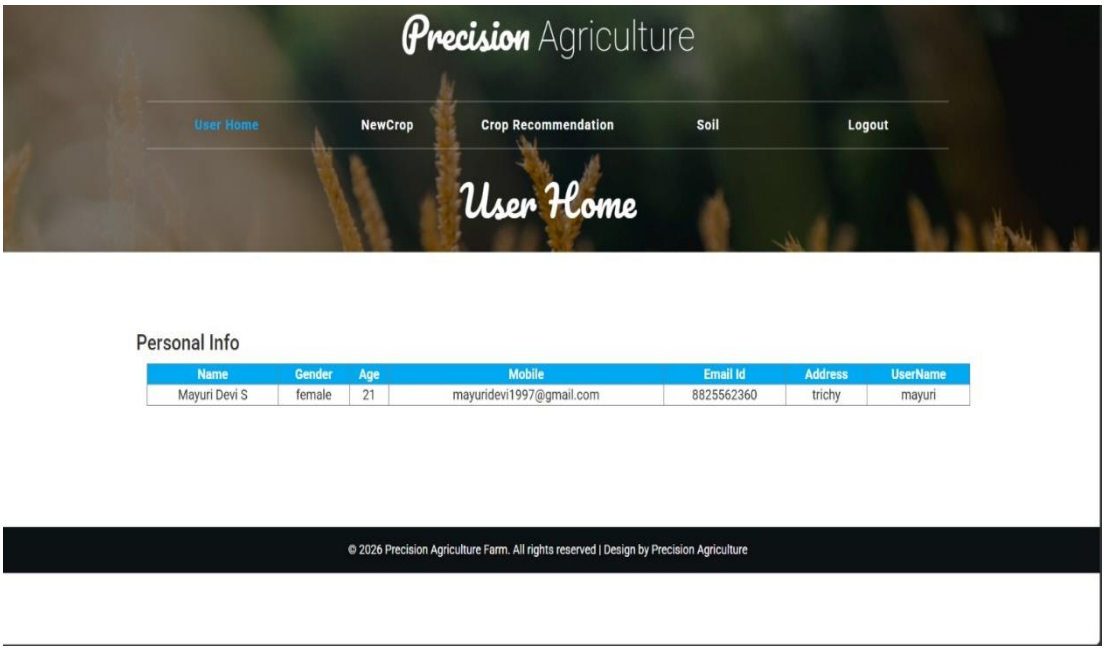

Figure B4. User Profile Page

Figure B5. Crop Data Entry Page

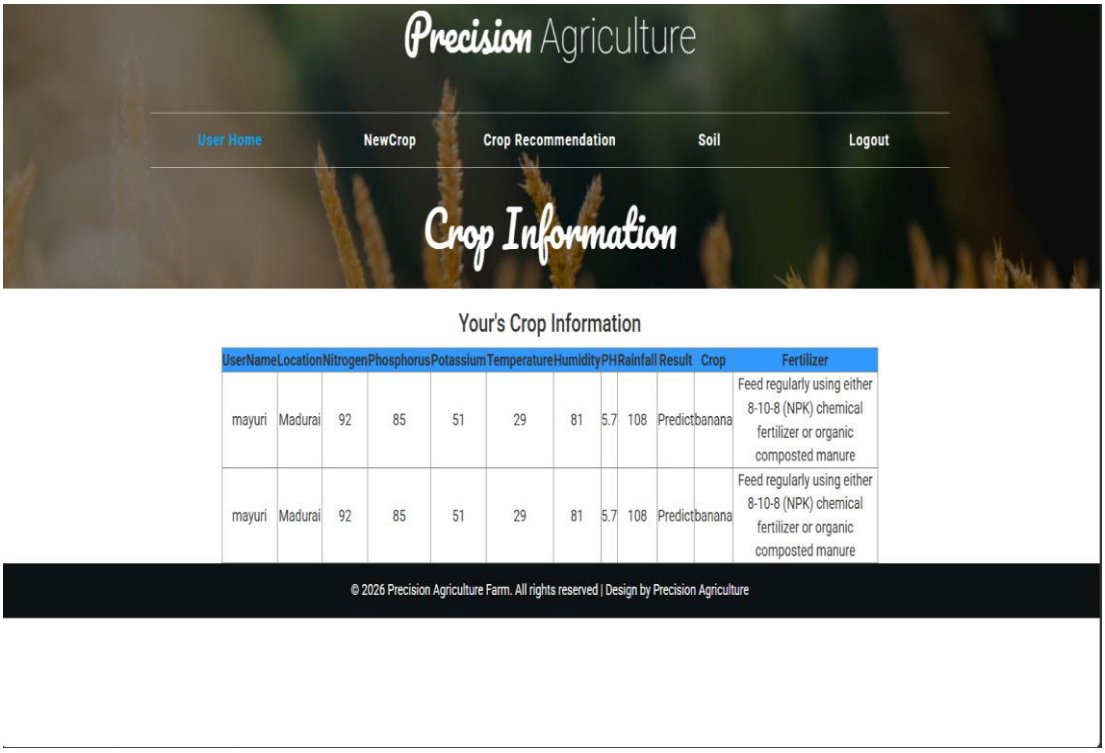

Figure B6. Crop And Fertilizer Recommendation Page

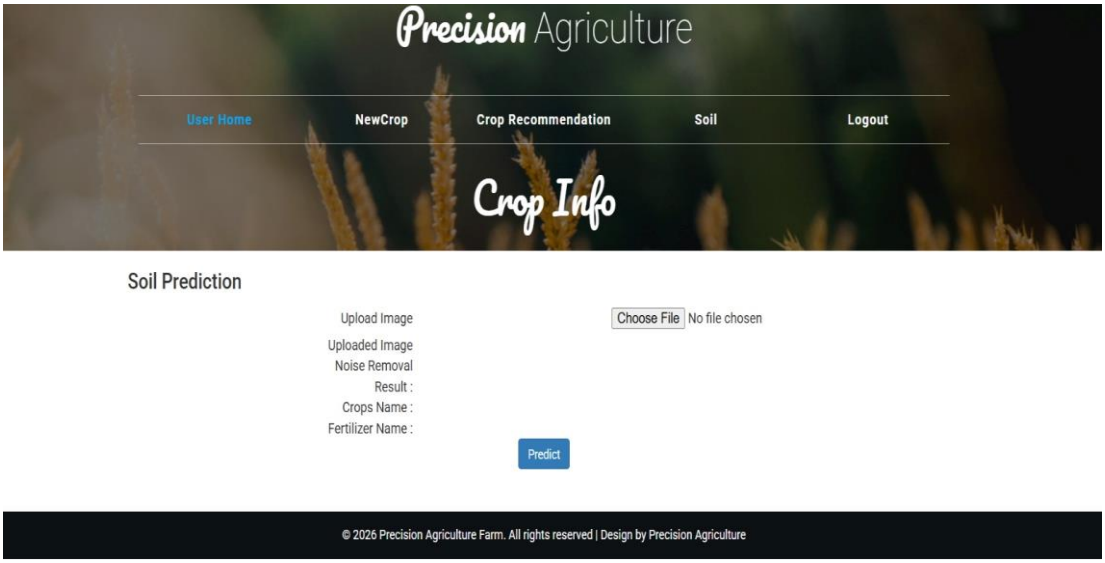

Figure B7. Soil Image Upload Page

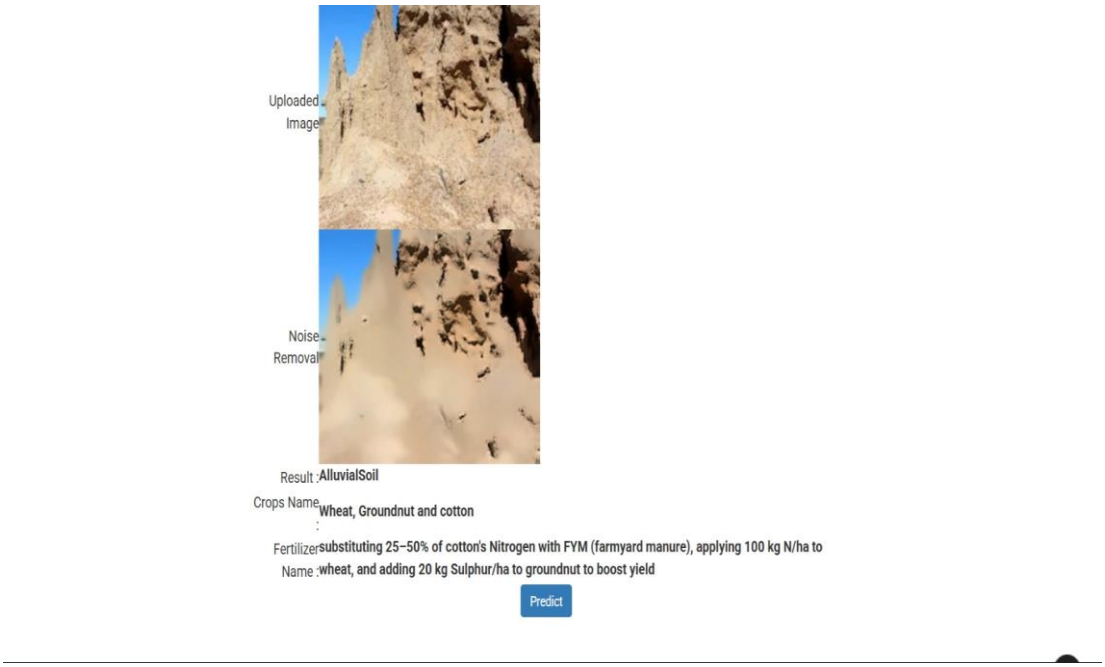

Figure B8. Soil Analysis Result Page

REFERENCES

1. R. Sujatha, G. Isakki, and S. Karthik, “Crop Recommendation System using Machine Learning,” International Journal of Engineering Research & Technology (IJERT), vol. 9, no. 5, pp. 123–126, 2020.

2. S. Navin Kumar, M. R. Prasad, and P. V. Nageswara Rao, “Smart Agriculture using IoT and Machine Learning,” International Journal of Recent Technology and Engineering (IJRTE), vol. 8, no. 6, pp. 2277–3878, 2020.

3. P. Revathi and M. Hemalatha, “Advance Computing Enrichment Evaluation of Cotton Leaf Spot Disease Detection using Image Edge Detection,” International Journal of Computer Applications, vol. 10, no. 6, pp. 1–6, 2012.

4. S. Mohanty, D. Hughes, and M. Salathé, “Using Deep Learning for Image-Based Plant Disease Detection,” Frontiers in Plant Science, vol. 7, pp. 1419, 2016.

5. J. R. Quinlan, “Induction of Decision Trees,” Machine Learning, vol. 1, no. 1, pp. 81–106, 1986.

6. L. Breiman, “Random Forests,” Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

7. Food and Agriculture Organization, “FAOSTAT Statistical Database,” 2023.

8.Kaggle, “Crop Recommendation Dataset,” Available: https://www.kaggle.com

9. TensorFlow, “TensorFlow: Large-Scale Machine Learning Library,” 2023.

10. OpenCV, “Open Source Computer Vision Library,” 2023.